



Data Analytics Technology

Examples DM

Catarina Silva, catarina@dei.uc.pt

Question 1

Indicate, with justification, which performance metric you would choose, in each case, to evaluate (i) classification algorithms and (ii) regression algorithms.

Question 2

Suppose you work for SayberMail company that offers data analytics solutions. The company must provide a solution in order to understand if users reply “reply” or “not reply” to a specific email from a user. To do this, it collected data from 10 volunteer users. As the information is sensitive, users' names and email addresses have been anonymized using an alphanumeric hash code such as "User35CB8E5", as shown in Figure 1.

- (a) Comment on the anonymization of data, indicating the consequences they have for a data analytics solution
- (b) Explain why the data analytics solution will require the existence of the 3 sets: training, validation and testing
- (c) For the solution to be provided by the company to the customer, which algorithm would you choose, justifying your choice, and indicate how you would proceed to implement the entire pipeline in your software of choice.

	Train	Validation	Test	User Total
User35CB8E5	1,995	2,005	657	4,657
UserCE3FDB4	1,067	1,067	356	2,490
User6AACED	1,827	1,822	600	4,249
User7E601F9	531	528	173	1,232
User68251CD	600	602	198	1,400
User223AECA	532	532	179	1,243
UserFF0F29E	2,202	2,199	729	5,130
User25C0488	1,181	1,182	393	2,756
User811E39F	1,574	1,565	513	3,652
User10628A6	485	485	163	1,133
Total	11,994	11,987	3,961	27,942
Average	1278.77778	1278	422	2978.77778

Figure 1 –Number of emails in training, validation and testing sets